Manikanta Reddy Kallam

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Professional Summary

- Possess over **5 years** of experience in Data Engineering, specializing in designing ETL pipelines and implementing efficient storage solutions for petabytes of data.
- Proficient in deploying fault-tolerant ETL solutions on **GCP and AWS** cloud environments.
- Developed a **4-layer architecture (raw, curated, semantic, and consumption)** to process data from raw information to meaningful insights.
- Designed data storage solutions including **SQL, NoSQL, and object storage databases** to ensure high availability and scalability.
- Utilized data warehousing resources such as **GCP BigQuery and AWS Redshift** to warehouse petabyte-scale data for insight dashboard and machine learning algorithms.
- Created a database system to store **transactional data (Postgres), metadata (MongoDB), secret manager (AWS Secret Manager), and object data (S3)**.
- Experienced in web scraping with tools like **Selenium and Beautifulsoup** to fetch data from multiple web pages.
- Demonstrated expertise in writing complex **SQL** queries to extract insights from large datasets.
- Implemented modern transformation tools like **Data Build Tool** to transform data effectively with reduced computing load.
- Skilled in writing **automation scripts** using Python for various tasks, including **monitoring daily inbound files** and automating production environment tasks.
- Developed **Google Data Studio-powered** insights to track the progress of development work on user stories.
- **Scheduled and initiated batch file** loading using orchestration tools like **Airflow** based on event triggers.
- Designed **Continuous Integration & Continuous Deployment** pipelines to automate the production deployment process, reducing manual efforts by 5 hours.
- Implemented **Secure Virtual Private Cloud** to transfer large amounts of data across multi-cloud environments.
- Deployed **notification tool via slack channel/email** to notify changes about subscriptions to users.
- Served as a **product designer** for a startup, **designing a secure web application with user registration, two-factor authentication, secure login, and payment functionalities**.
- Developed **RESTful APIs** to securely interact with backend databases using **JWT token-based authentication**.
- Actively involved in **providing proof of concepts (POC)** for various data transformation and ingestion use cases.
- Experienced in working with both **waterfall and agile** project management methodologies.
- Worked on wide service management tools like **JIRA, and Service-now** for change requests, incidents, and problem ticket management and created documentation for production handoff.
- Responsible for creating and maintaining **Kanban boards** to facilitate efficient project outcomes.
- Led cross-team functional collaboration projects from gathering requirements to **designing interface mapping documents, setting up data ingestion FTP pipelines, ETL development, QA testing, Production release, and active maintenance**.
- **Led a team of 5 members** to manage production environment issues and meet client requirements effectively.
- **Mentored a team of interns** in creating three internal dashboards, collaborating closely with marketing account managers to streamline manual processes by 75%.
- Enthusiastic about learning and implementing new enterprise tools and technologies.
- **Hardworking and self-driven individual with strong analytical, problem-solving and excellent communication skills**.

Professional Experience

- **PaloAlto soft** Remote
  Data Engineer *September 2022 - July 2024*
  - **Responsibilities**: To design backend RESTful & GraphQL API's for secure data retrieval to the frontend application
  - **Frameworks & Services**: AWS, Python, serverless framework, lambda, S3, MongoDB, API Gateway, Cognito, Code Pipeline, Amplify and PostgreSQL.
  - **Goals Achieved**:
    * Designed, developed, deployed, and tested a comprehensive web application with features including user registration, secure login, two-factor authentication, and payment portal integration.
    * Implemented robust security measures to ensure user data protection and application integrity, including the development of a secure login system and two-factor authentication using AWS Cognito.
    * Integrated a seamless payment portal with Stripe, enabling smooth and secure transactions for users and enhancing the overall functionality of the web application.
    * Managed the entire product development lifecycle from concept to deployment, ensuring timely delivery and adherence to project specifications.
    * Collaborated closely with designers, stakeholders, and other developers to align the product with user needs and business goals, utilizing agile development practices to iterate quickly and respond to feedback.
    * Enabled CI/CD pipelines with AWS code pipeline to automate the deployment process which reduces manual effort by 3 hours.
    * Constructed Relational Databases to store transactional details about users.
    * Facilitated the management of electronic or digital identities using AWS IAM to programmatic and physical users.
    * Responsible for interpreting data requirements to customers to fulfill requirements and needs.

- **Infosys Limited - Loreal USA** Hybrid
  Senior Systems Engineer - Data Engineering *February 2021 - July 2022*
  - **Responsibilities**: To enable data modeling and multigrain insights on sales and ads running on the e-commerce portal
  - **Frameworks & Services**: Google Cloud Platform, Python, DBT, Composer-Airflow, Flask, Cloud Function, Cloud Run, Bigquery, Firestore, DataFlow.
  - **Goals Achieved**:
    * Built High scalable 4-layer architecture(raw, curated, semantic & Consumption) to handle huge workloads effectively
    * Developed and maintained data pipelines and analytics solutions on GCP for data processing and warehousing efficiently.
    * Created data models and implemented automated data validation checks to ensure data quality.
    * Managed and optimized Firestore databases to ingest metadata for inbound files, supporting real-time tracking and validation to avoid error rate by 30%.
    * Worked with SQL and NoSQL databases to query datasets and analyze data.
    * Created data scraping tool with Flask and Pandas to scrape data across websites which reduced manual efforts by 70%.
    * Achieved metadata-driven composer(Airflow) to orchestrate data pipelines to perform ETL operations.
    * Leveraged DBT (Data Build Tool) for data transformation, enabling modular and maintainable SQL codebases.
    * Created automated scripts to monitor daily file loading in cloud storage improving efficiency and reducing processing time by 60% for internal reporting and interpreting
    * Collaborated with cross-functional teams to gather requirements and deliver data solutions that meet business needs.
    * Worked with business stakeholders/clients to identify and understand key performance indicators, and created data models to support business needs.

- **Infosys Limited** Hybrid
  Systems Engineer - Data and Analytics *February 2020 - February 2021*
  - **Responsibilities**: Generating Daily Report of Production Environment Status and Fulfillment of Adhoc Client Requirements
  - **Frameworks & Services**: Data Flow, Cloud functions, Airflow, Bigquery, Tableau, Google data Studio & CI/CD
  - **Goals Achieved**:
    * Implemented Python-based automation to generate daily inbound file status reports, streamlining the process and significantly reducing manual effort by 90%. These automated reports were delivered promptly to the intended audience, enhancing operational efficiency and enabling stakeholders to make informed decisions based on up-to-date information.
    * Timely fulfillment of ad-hoc client requirements led to a 10% increase in profits.
    * Developed CI/CD pipelines using codebuild and github to automate code deployment to avoid manual intervention and increasing security compliance.
    * Worked on Agile environment by prioritizing customer feedback and involvement throughout the development process.
    * Designed and implemented Tableau dashboards to analyze and visualize revenue generation from advertising campaigns for an e-commerce organization. These dashboards provided actionable insights into ad performance, facilitating data-driven decision-making and optimizing revenue strategies.

- **Paloalto Soft** Remote
  Software Developer *March 2019 - January 2020*
  - **Responsibilities**: Development of smart home automation devices based on real-time sensors and user control
  - **Frameworks & Services**: AWS, Pub/Sub,lambda, API Gateway,EC2 instances, Embedded C, Arduino, Python and SQL.
  - **Goals Achieved**:
    * Designed, developed, and maintained firmware for home automation devices and complex systems, including smart locks, thermostats, and security systems.
    * Constructed relational databases to store transactional data for driving multiple user-defined scenarios.
    * Designed and developed firmware for smart home lighting systems, including drivers for various lighting components and integration with Real-time data and voice assistants such as Amazon Alexa and Google Home.
    * Worked on CI/CD pipelines with AWS cloud build to automate deployment process which saves 90% of production deployment time.
    * Deployed managed ec2 instances to enable multiple low-cost cloud storage systems which reduced 60% of the cost.
    * Created REST API's with AWS lambda and API gateway to enable serverless architecture.
    * Implemented Graph databases to provide faster data operations on complex relational data.
    * Worked with wireless communication protocols such as Wi-Fi, Zigbee, and Bluetooth, to ensure seamless connectivity between devices and mobile applications.

Skills Summary

Languages	: Python, SQL, C++, Java, Embedded C
Frameworks	: Data Build Tool (DBT), Flask, pyspark, streamlit
Tools	: Docker, Jira, git, Apache Airflow
Databases	: PostgreSQL, SQL server, Bigquery, Redshift, Snowflake
Platforms	: Google Cloud Platform, AWS, Linux
Reporting Tools	: Google Data Studio, Tableau, PowerBI
Soft Skills	: Decision Making, Problem Solving, Team management, Innovative, Technical writing.

Education

- **California State University, Eastbay** Hayward, CA
  *Master of Science in Statistics and Data Science - **3.75/4 GPA*** *August 2022 - April 2024*
- **Jawaharlal Nehru Technological University** Kakinada, India
  *Bachelor of Technology in Electronics and Computer Engineering - **3/4 GPA*** *July 2015 - April 2019*

Projects

- **CNN-based Brain tumor detection with Image segmentation**: (Work in progress) Research-oriented, open-source medical analysis application specializing in brain tumor detection using semantic image segmentation techniques. The tool enables quick and accurate analysis, benefiting medical practitioners by expediting diagnosis timelines and conserving resources. Its open-source nature fosters collaboration and innovation within the medical community, representing a significant stride towards leveraging technology to improve healthcare outcomes.
  Tech: Python, OpenCV, DeepLabV3 TensorFlow, Keras.
- **Cloud-based Serverless ETL architecture**: (Work in progress) The Serverless model gives us high flexibility in scaling and maintaining workload by reducing 60% of the overall cost for an ETL/ELT data pipeline. With this open-source application architecture, one can deploy on any cloud with their respective services with ease and comfort.
  Tech: Python, GCP cloud, AWS cloud, DBT, FLASK, docker
- **SanFrancisco Crime Data Analytics**: A Tableau-powered analytics dashboard for public crime data, offering citywide crime insights and status summaries. Utilized Etl pipelines, and visualizations including maps and charts, to convey key crime trends and patterns, facilitating data-driven decision-making.
  Tech: Tableau, Bigquery, DBT, RESTful API, Flask, Apache Airflow (September '23)